TEXORA · LECTURE SHEET

Water Pollution: Effluent in Wet Processing and Its Treatment

The character of textile effluent, the parameters that describe it, physical, chemical and biological (aerobic) treatment, electrochemical colour removal, ETP sizing with a full worked example, and disposal systems.

TE-4204 · ENVIRONMENTAL ENGINEERING AND POLLUTION CONTROL · TOPIC 6 · LEVEL 4 TERM 2

1. Water pollution and the wet-processing effluent

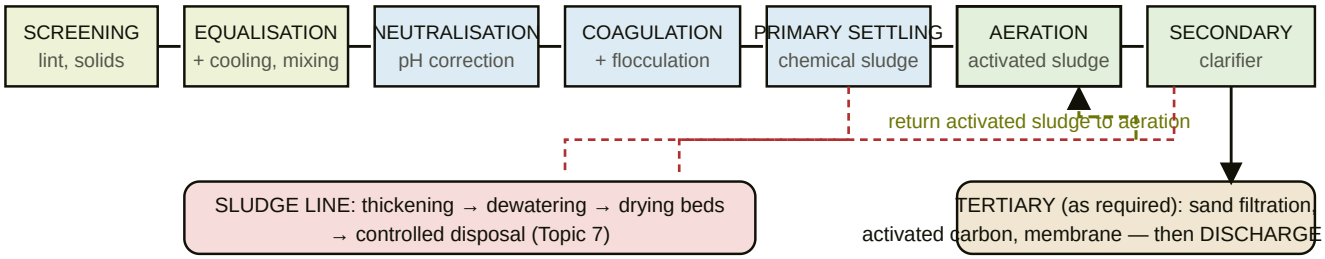
Water pollution is the degradation of a water body by added substances or heat until its legitimate uses — drinking, irrigation, fisheries, bathing, industry — are impaired. Wet processing is the textile industry's great water user and its great polluter: a dyeing and finishing plant may use on the order of a hundred litres of water per kilogram of fabric across all its operations, and returns most of it as effluent carrying the chemicals of Topic 3. The receiving water suffers in specific, measurable ways: microbes oxidising the organic load consume dissolved oxygen (fish suffocate when oxygen crashes); colour blocks light and photosynthesis; salts raise salinity beyond irrigation tolerance; and toxic residues act directly on aquatic life and on people using the water downstream.

PARAMETER	MEANING	WHY IT IS CONTROLLED
pH	Acidity/alkalinity of the effluent	Aquatic life needs near-neutral water; extremes corrode and kill
BOD_5	Biochemical oxygen demand — oxygen microbes consume oxidising the biodegradable load in 5 days	Predicts oxygen depletion in the receiving water
COD	Chemical oxygen demand — oxygen equivalent of everything chemically oxidisable	Total oxidisable load; COD ≥ BOD always; the gap shows how much is non-biodegradable
TSS	Total suspended solids	Siltation, turbidity, sludge deposits
TDS	Total dissolved solids (mainly dyeing salts)	Salinity for irrigation and drinking use; very hard to remove

Colour	Residual dye, measured as absorbance or Pt–Co units	Visible at low concentration; blocks light; public's first evidence of pollution
Oil & grease, temperature, specific toxics	Process-specific	Surface films, oxygen stress, direct toxicity

2. The treatment train

An effluent treatment plant (ETP) is a sequence of unit operations, each removing what the next cannot handle. The standard train for textile effluent:



Physical → chemical → biological: each stage removes the fraction the next stage cannot.

Figure 1 — A typical textile ETP train. Equalisation first (textile effluent varies hour by hour), chemical treatment for colour and solids, aerobic biological treatment for BOD, tertiary polishing as required, and a sludge line that must never be forgotten.

3. Physical treatment

Screening removes lint, fibre and coarse solids on bar or rotary screens, protecting pumps and downstream units. **Equalisation** is the textile ETP's most important tank: flow and strength vary enormously between a dye-bath dump and a rinse period, so effluent is held and mixed (often with cooling and pre-aeration) to feed the plant a steady, average stream — nearly every treatment failure story begins with a missing or undersized equalisation tank. **Sedimentation** settles suspended solids by gravity in clarifiers; **flotation** (dissolved-air flotation) lifts light solids, fibres and grease with fine bubbles; **filtration** (sand, multimedia, membranes) polishes suspended matter out of treated water, especially before reuse.

4. Chemical treatment

Neutralisation corrects pH with acid or alkali so that later stages (especially biological ones) can work. **Coagulation–flocculation** is the workhorse for colour and colloidal solids: a coagulant such as alum or ferric salt neutralises the charges that keep fine particles and many dyes dispersed, and a polymer flocculant builds large, settleable flocs; jar tests on the actual effluent set the dose, because dyehouse chemistry changes daily. **Chemical oxidation** (with ozone, peroxide-based or chlorine agents) attacks colour and refractory organics; **precipitation** removes metals as hydroxides. Chemical treatment is fast and compact but buys its performance with reagent cost and with chemical sludge — the trade-off that the costing of Topic 7 makes visible.

5. Biological (aerobic biochemical) treatment

Biodegradable organics — the BOD — are removed by micro-organisms that eat them. In the **activated sludge process**, effluent is aerated in a basin with a dense microbial culture (the "activated sludge", measured as MLSS — mixed liquor suspended solids); the microbes oxidise organics to CO_2, water and new cell mass; a secondary clarifier then separates the culture, most of which is returned to the basin. The design quantities are the **organic load** (kg BOD/day), the **hydraulic retention time (HRT)**, and the **food-to-microorganism ratio (F/M)**. Alternatives and complements include trickling (bio) filters, rotating biological contactors, aerated lagoons and, for compact high-rate treatment, membrane bioreactors; anaerobic pre-treatment is often used ahead of aerobic stages for very strong effluents, biogas being a bonus. Aerobic treatment is cheap per kilogram of BOD removed but needs steady feeding (equalisation again), nutrients in proportion, and protection from toxic shock loads — which is why concentrated chemical streams are segregated (Topic 3).

WORKED EXAMPLE — EFFLUENT LOAD AND AERATION-BASIN SIZING

A dyeing plant discharges 600 m³/day of effluent with BOD_5 320 mg/L and COD 900 mg/L. (Note: 1 mg/L = 1 g/m³.)

BOD load = 600 × 320 ÷ 1,000 = **192 kg BOD/day**. COD load = 600 × 900 ÷ 1,000 = **540 kg COD/day**.

(The COD:BOD ratio of ≈ 2.8 warns that a fair share of the load is poorly biodegradable — expect chemical treatment to carry part of the removal.)

Aeration basin for an HRT of 8 hours: flow = 600 ÷ 24 = 25 m³/h → volume = 25 × 8 = **200 m³**.

At MLSS 3,000 mg/L (= 3 kg/m³), the basin holds 200 × 3 = 600 kg of microbial solids, so F/M = 192 ÷ 600 = **0.32 kg BOD per kg MLSS per day** — inside the usual design band for conventional activated sludge.

If the plant achieves 92 % BOD removal overall, outlet BOD = 320 × 0.08 = **25.6 mg/L**.

Lesson: loads are masses, not concentrations. Doubling production at the same concentration doubles the kilograms the biology must eat — dilution never reduces a load.

6. Electrochemical technology for colour removal

Electrocoagulation (EC) generates coagulant in place: a sacrificial iron or aluminium anode dissolves under direct current, releasing metal ions that hydrolyse into the same hydroxide flocs a chemical coagulant would give, while hydrogen bubbles at the cathode float flocs upward. It is effective on colour, dispersed and vat dyes, oils and some metals, needs no stored coagulant chemicals, and produces a compact unit suited to small and variable flows. **Electro-oxidation** instead destroys organics at the anode surface or through electrogenerated oxidants. The governing law for EC dosing is Faraday's:

$$\text{Metal dissolved (mol)} = (I \times t \div F) \div \text{electrons per ion}, \quad F = 96{,}485 \text{ C/mol}$$

WORKED EXAMPLE — IRON DOSE FROM AN ELECTROCOAGULATION CELL

A cell runs at 100 A for 1 hour with iron anodes ($Fe \rightarrow Fe^{2+} + 2e^-$).

Charge passed = 100 × 3,600 = 360,000 C → moles of electrons = 360,000 ÷ 96,485 = 3.73 mol.

Iron dissolved = 3.73 ÷ 2 = 1.87 mol × 55.85 g/mol ≈ **104 g of iron per hour**.

This is the theoretical dose; real cells dose a little less (current efficiency below 100 %), and the operator trades dose (current) against colour removal and electrode wear.

Limits of electrochemical methods: electrode consumption and passivation, electricity cost per cubic metre (which rises with the effluent's conductivity demand and target quality), and the metal hydroxide sludge still to be disposed of. They are best used as targeted colour/polishing units or for small factories, complementing rather than replacing the main train.

7. Disposal systems and reuse

Treated effluent is disposed of by **discharge to surface water** (the commonest route — the discharge must meet the consent limits for that water body's use), **discharge to a public sewer** (where the municipal plant accepts it, usually with pretreatment conditions and strength charges), **land application** (only for suitable quality, under soil and groundwater safeguards), or **reuse inside the factory** — the direction of travel everywhere: treated water returns to washing, floor cleaning, gardening and, after membrane polishing, to process use. At the extreme stands **zero liquid discharge (ZLD)**: evaporation/crystallisation of the final brine so no liquid leaves the site — technically possible, but energy-hungry and leaving a solid salt residue; it is adopted where water or consent conditions force it. Sludge from every route goes to the solid-waste system of Topic 7. Monitoring closes the loop: flow-proportional sampling of the outlet for pH, BOD, COD, TSS, TDS and colour, logged against the consent.

Glossary

BOD_5 — oxygen consumed by microbes oxidising the biodegradable load in 5 days at 20 °C. **COD** — oxygen equivalent of all chemically oxidisable matter. **MLSS** — mixed liquor suspended solids, the microbial mass concentration in the aeration basin. **HRT** — hydraulic retention time, basin volume ÷ flow. **F/M ratio** — food-to-microorganism ratio, kg BOD applied per kg MLSS per day. **Coagulation / flocculation** — charge neutralisation of fine particles / growth of settleable flocs. **Electrocoagulation** — generation of metal-hydroxide coagulant by dissolving a sacrificial anode. **Equalisation** — holding and mixing effluent to smooth flow and strength. **ZLD** — zero liquid discharge. **Consent limits** — the discharge conditions set in the plant's environmental clearance.

Key points

• Textile effluent is described by pH, BOD, COD, TSS, TDS and colour; BOD measures the biodegradable part, COD the total oxidisable load.

• The standard train runs physical → chemical → biological, with equalisation first and tertiary polishing as needed.

• Worked loads: 600 m^3/day at BOD 320 mg/L carries 192 kg BOD/day; an 8-hour HRT needs 200 m^3 of aeration volume; F/M = 0.32 at MLSS 3,000 mg/L; 92 % removal gives 25.6 mg/L out.

• Electrocoagulation doses coagulant by Faraday's law — 100 A for 1 h dissolves ≈ 104 g of iron — and suits colour removal and small flows.

• Disposal routes: surface water, sewer, land, internal reuse, or ZLD; every route leaves sludge to Topic 7 and monitoring to Topic 8.

Practice

1. An effluent flows at 800 m^3/day with BOD 400 mg/L. Compute the BOD load. What aeration volume gives a 10-hour HRT, and what overall removal percentage is needed to discharge at 50 mg/L?
2. Why must equalisation come before biological treatment in a dyehouse ETP? Name two failure symptoms when it is undersized.
3. An effluent has COD 1,200 mg/L and BOD 300 mg/L. What does the ratio tell you about treatability, and which treatment stage will carry the heavier burden?
4. Using Faraday's law, estimate the iron dissolved by an EC cell at 250 A in 2 hours (use the worked example's result and scale it).
5. Compare discharge to surface water, discharge to sewer and full reuse under four headings: pretreatment needed, cost driver, residual problem, best suited factory.

Answer key

1. Load = 800 × 400 ÷ 1,000 = **320 kg BOD/day**. Flow = 800 ÷ 24 = 33.3 m³/h → volume = 33.3 × 10 ≈ **333 m³**. Required removal = (400 − 50) ÷ 400 = **87.5 %**.
2. Dyehouse discharge swings between concentrated bath dumps and dilute rinses; biology needs a steady food supply, steady pH and no toxic shocks. Equalisation averages flow and strength (and cools the stream), so downstream units are sized for the average, not the peak. Symptoms of undersizing (any two): pH swings killing the biomass; colour/BOD breakthrough after each bath dump; clarifier solids washout at peak flow; frequent consent failures despite a "correctly sized" aeration basin.
3. COD:BOD = 4:1, a high ratio: only about a quarter of the oxidisable load is readily biodegradable, so biological treatment alone will leave much of the COD. Chemical treatment (coagulation/oxidation) and possibly electrochemical polishing must carry the heavier burden; source reduction of refractory chemicals should also be reviewed.
4. The worked example gives 104 g Fe for 100 A × 1 h = 100 A·h. Here charge = 250 × 2 = 500 A·h = 5 × 100 A·h → iron dissolved ≈ 5 × 104 = **520 g** (theoretical, before current efficiency).
5. Surface water: full treatment to consent limits needed; cost driver is the complete ETP; residual problem is sludge and any TDS the river must assimilate; suits most factories with a compliant ETP. Sewer: pretreatment to protect the sewer and municipal plant (screening, neutralisation, strength limits); cost driver is sewerage/strength charges; residual problem shifts load to the municipality; suits factories in serviced industrial estates. Full reuse: highest treatment (membrane polishing); cost driver is energy and membranes; residual problem is the concentrated reject stream; suits water-scarce sites or strict-consent zones, often combined with partial discharge.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib